Lab Report: Configuring IPv4 Static and Default Routes

Lab 2.2.2.5

Author: Aymane Es-salehy
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This report documents the configuration and verification of IPv4 static and default routes in a two-router network topology. The lab demonstrates the practical application of recursive static routes, directly connected static routes, and default routes to establish full network connectivity. Connectivity issues caused by missing routes are systematically identified and resolved.

1 Network Topology Overview

The initial phase involved checking the IPv4 address configurations on PC-A, PC-C, and the routers to establish baseline parameters.

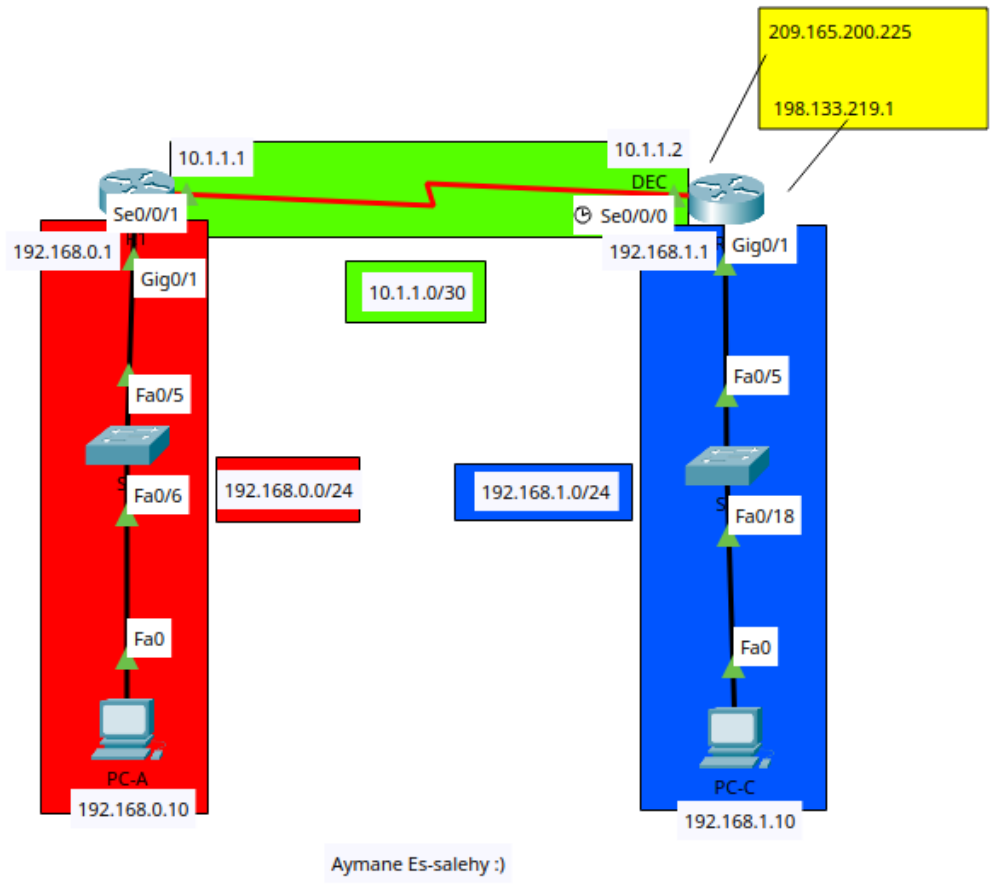


Figure 1: Final Network Topology in Cisco Packet Tracer.

The network consists of two routers ($R1, R3$) connected via a point-to-point serial link, with local area networks attached to each router. R3 also includes two loopback interfaces for testing purposes.

Device	Interface	IP Address	Mask	Default Gateway
R1	G0/1	192.168.0.1	255.255.255.0	N/A
R1	S0/0/1	10.1.1.1	255.255.255.252	N/A
R3	G0/1	192.168.1.1	255.255.255.0	N/A
R3	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A
R3	Lo0	209.165.200.225	255.255.255.224	N/A
R3	Lo1	198.133.219.1	255.255.255.0	N/A
PC-A	NIC	192.168.0.10	255.255.255.0	192.168.0.1
PC-C	NIC	192.168.1.10	255.255.255.0	192.168.1.1

Table 1: IP Addressing Table.

2 Basic Configuration

subsectionPC Configuration Both PCs were configured with their respective IP addresses, subnet masks, and default gateways according to the Addressing Table.

Device	IP Address	Subnet Mask	Default Gateway
PC-A	192.168.0.10	255.255.255.0	192.168.0.1
PC-C	192.168.1.10	255.255.255.0	192.168.1.1

Table 2: PC Configuration Settings.

2.1 Router R1 Interface Configuration

R1 was configured with hostname and interface IP addresses. The GigabitEthernet0/1 interface connects to the LAN (PC-A network), while Serial0/0/1 connects to R3.

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

! Configure LAN interface
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ip address 192.168.0.1 255.255.255.0
R1(config-if)# no shutdown

! Configure WAN interface
R1(config)# interface Serial0/0/1
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# no shutdown
```

Listing 1: R1 Interface Configuration

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/0/1
 ip address 10.1.1.1 255.255.255.252
!
```

Figure 2: R1 Running Configuration.

2.2 Router R3 Interface Configuration

R3 was configured with hostname, interface IP addresses, and loopback interfaces. The GigabitEthernet0/1 interface connects to the LAN (PC-C network), while Serial0/0/0 connects to R1.

```
Router> enable
Router# configure terminal
Router(config)# hostname R3

! Configure LAN interface
R3(config)# interface GigabitEthernet0/1
R3(config-if)# ip address 192.168.1.1 255.255.255.0
R3(config-if)# no shutdown

! Configure WAN interface
R3(config)# interface Serial0/0/0
R3(config-if)# ip address 10.1.1.2 255.255.255.252
R3(config-if)# clock rate 128000    (Required because R3 is the DCE
    device)
R3(config-if)# no shutdown

! Configure Loopback 0
R3(config)# interface Loopback0
R3(config-if)# ip address 209.165.200.225 255.255.255.224

! Configure Loopback 1
R3(config)# interface Loopback1
R3(config-if)# ip address 198.133.219.1 255.255.255.0
```

Listing 2: R3 Interface Configuration

```
interface Loopback0
 ip address 209.165.200.225 255.255.255.224
!
interface Loopback1
 ip address 198.133.219.1 255.255.255.0
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.1.1.2 255.255.255.252
 clock rate 128000
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
```

Figure 3: R3 Running Configuration.

2.3 Common Security Settings (Applied to Both Routers)

For device management and security purposes, the following settings were applied to both R1 and R3:

```
! Disable DNS lookup to prevent delays on mistyped commands
no ip domain-lookup

! Set privileged EXEC password
enable password lm02s6

! Configure console access
line console 0
 password lm02s6
 login

! Configure VTY (Telnet/SSH) access
line vty 0 4
 password lm02s6
 login
```

Listing 3: Common Security Settings for R1 and R3

Command	Explanation
`clock rate 128000`	Required on DCE serial interfaces to provide clocking synchronization. Without this, the serial link will not come up. The value 128000 represents 128 kbps.
`no ip domain-lookup`	Prevents the router from attempting DNS lookups for mistyped commands, which can cause delays.
`enable password`	Sets the password required to enter privileged EXEC mode (#).
`line console 0`	Configures the console port for local device access.
`line vty 0 4`	Configures virtual terminal lines for remote access (Telnet/SSH).
`login`	Requires password authentication before granting access.

Table 3: Key Configuration Commands Explanation.

3 Initial Connectivity Testing

3.1 Test Results

Before configuring static routes, connectivity was tested between various devices.

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC-A	R1	ICMP	
	Successful	PC-C	R3	ICMP	
	Successful	R1	R3	ICMP	
	Failed	PC-A	PC-C	ICMP	

Figure 4: Initial Connectivity Test Results.

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.133.219.1

Pinging 198.133.219.1 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Request timed out.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 198.133.219.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 5: PC-A Ping to Loopback Interfaces (Failed).

3.2 Analysis

PC-A cannot reach PC-C or the loopback interfaces because R1 does not have routes to these networks in its routing table. While PC-C can reach the loopbacks (they're on the same router), communication between different networks requires proper routing configuration.

4 Routing Table Analysis

4.1 R1 Initial Routing Table

The initial routing table on R1 shows only directly connected networks:

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/1
L       10.1.1.1/32 is directly connected, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/1
L       192.168.0.1/32 is directly connected, GigabitEthernet0/1
```

Figure 6: R1 Initial Routing Table.

Networks Missing from R1:

- 192.168.1.0/24 (PC-C network)
- 198.133.219.0/24 (Loopback 1)
- 209.165.200.224/27 (Loopback 0)

4.2 R3 Initial Routing Table

R3's routing table shows directly connected networks including the loopback interfaces:

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/0
L       10.1.1.2/32 is directly connected, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
     198.133.219.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.133.219.0/24 is directly connected, Loopback1
L       198.133.219.1/32 is directly connected, Loopback1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/27 is directly connected, Loopback0
L       209.165.200.225/32 is directly connected, Loopback0

R3#
```

Figure 7: R3 Initial Routing Table.

Networks Missing from R3:

- 192.168.0.0/24 (PC-A network)

5 Static Route Configuration

5.1 Directly Connected Static Route (R3 → 192.168.0.0/24)

A directly connected static route specifies the exit interface, requiring only one lookup.

```
R3(config)# ip route 192.168.0.0 255.255.255.0 Serial0/0/0
```

Listing 4: Directly Connected Static Route on R3

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/0
L       10.1.1.2/32 is directly connected, Serial0/0/0
S    192.168.0.0/24 is directly connected, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
     198.133.219.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.133.219.0/24 is directly connected, Loopback1
L       198.133.219.1/32 is directly connected, Loopback1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/27 is directly connected, Loopback0
L       209.165.200.225/32 is directly connected, Loopback0

R3#
```

Figure 8: R3 Routing Table After Directly Connected Static Route.

Note: Only this single static route was configured on R3. Instead of configuring multiple static routes on R1 to reach PC-C and the loopback networks, a default route will be used on R1 (see Section 6).

6 Default Route Configuration

6.1 Understanding Default Routes

A default route is a static route that matches any destination IP address. It acts as a "catch-all" for packets that don't match any other route.

- **Destination:** 0.0.0.0 (any network)
- **Subnet Mask:** 0.0.0.0 (any mask)
- **Use:** Gateway of last resort

6.2 Configuring a Default Route on R1

Instead of configuring individual static routes for each network on R3, a single default route was used on R1 to reach all unknown networks via R3.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 10.1.1.2
```

Listing 5: Default Route Configuration on R1

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/1
L       10.1.1.1/32 is directly connected, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/1
L       192.168.0.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 is directly connected, Serial0/0/1
```

Figure 9: R1 Routing Table After Default Route Configuration.

Why this approach is better:

- Only one default route needed on R1 instead of three separate static routes
- If new networks are added to R3, R1 will automatically reach them
- Simpler configuration and less maintenance

7 Final Verification

After applying all configuration changes, end-to-end connectivity tests were performed.

7.1 PC-A Ping Results

```
Pinging 198.133.219.1 with 32 bytes of data:

Reply from 198.133.219.1: bytes=32 time=34ms TTL=254
Reply from 198.133.219.1: bytes=32 time=11ms TTL=254
Reply from 198.133.219.1: bytes=32 time=29ms TTL=254
Reply from 198.133.219.1: bytes=32 time=29ms TTL=254

Ping statistics for 198.133.219.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 34ms, Average = 25ms

C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=15ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=17ms TTL=254
Reply from 209.165.200.225: bytes=32 time=14ms TTL=254

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 17ms, Average = 11ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=18ms TTL=126
Reply from 192.168.1.10: bytes=32 time=10ms TTL=126
Reply from 192.168.1.10: bytes=32 time=12ms TTL=126
Reply from 192.168.1.10: bytes=32 time=19ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 19ms, Average = 14ms
```

Figure 10: Final Connectivity Verification (100% Success).

8 Conclusion

This lab successfully demonstrated the configuration and verification of IPv4 static and default routes. The key accomplishments included:

1. **Basic Configuration:** Configured hostnames, interface IP addresses, security settings, and the required clock rate on the DCE serial interface.
2. **Routing Table Analysis:** Identified missing routes on both routers. R1 lacked routes to 192.168.1.0/24, 209.165.200.224/27, and 198.133.219.0/24, while R3 lacked a route to 192.168.0.0/24.
3. **Static Route Implementation:** Configured a directly connected static route on R3 to reach PC-A's network.
4. **Default Route Implementation:** Configured a single default route (0.0.0.0/0) on R1 instead of multiple static routes, demonstrating a more scalable and maintainable approach for stub routers.

All configurations were verified through successful ping tests, confirming full IPv4 connectivity across the network topology.